Staff Days — Week 2, Day 8 (Wed Aug 5)

> **Format note:** Joe's session ran the entire morning (~2h50m) and it's the densest academics content of the two weeks. No post-lunch tee-up was recorded — the afternoon was campus build time. This was also Joe's last staff days appearance; he flew out for the Huberman podcast.

The exercise — *One year from now, why didn't we deliver?*

`0:00–0:35`

The framing: a year from now we sit in this room and ask whether we delivered the three commitments to every kid across 50+ campuses. There will be new mistakes — that's fine. What we can't say is that we failed on something **we already knew about last year and didn't fix.** So: put yourself a year forward and name the reasons.

What the room named

Kids not leveling up · parents losing faith in the model · love of school slipping · test failure rate staying high · enrolling students without parent alignment · kids not prepared for the next step, especially middle→high school · parents shielding kids from our standards · contentious guide–student relationships · not supporting neurodivergent students · not hitting 2X in math · parents and students not accepting hole-filling ("go slow to go fast") · workshops that are stunts rather than real life skills, or simply not rigorous enough · having to do reputation-repair PR instead of building one · guide transitions · Alpha Canon not actually producing consistency · minority students and staff feeling supported · high-performing middle schoolers hitting a content cliff · students we shouldn't have admitted · the guide-to-student ratio increase · StudyFilm not working · being told three weeks before MAP that the XP bar just jumped.

The one Joe expanded on: returning vs. new students

A first-year family compares Alpha to their old school, so almost anything lands. **In year two, their peak from last year becomes the new floor.** That's true academically (new students get the easy catch-up 2X; returning students don't), in life skills (last year's best workshop is now the baseline), and in guides — every guide gets compared to the best guide that kid ever had. One campus reported that **none of their returning first graders hit 2X**.

"We're competing with ourselves." Joe's view is this is mostly good: the students furthest into the system have the highest standards. Alpha High seniors interview guide candidates and open with "I'm going to make a million dollars this year, how are you going to help me?"

Painfully insightful metrics

The test: if you could never visit a campus and only had a dashboard, what would you need to see? Online, that metric is **percent cheating** — and it's hard to get. The example: an Ivy League professor whose class aced the midterm and failed the in-person final, because he never had the metric telling him who was actually doing the work. The second example: a classical school assigning great books whose SAT scores make it impossible those books were read. **"Your kids aren't reading them. You're assigning them."** Assigning the work is not evidence of learning.

The academic model, gate by gate

`0:35–2:10`

The chain, working backwards from the goal:

daily lesson → unit quiz → grade-level state test → January MAP → year-end MAP (2X)

Each is a mastery gate, and each has known failure points.

Mastery is the secret sauce

"If you had to say what the secret sauce of our system was — that's it." TimeBack is the only mastery-enforcing platform. Joe's read of Bloom's Two Sigma: everyone cites the *individualized tutor*, but the finding is a **mastery-based** individualized tutor. That's precisely why chatbots don't work.

The market dynamic: vendors who understand mastery lower the bar to sell. Khan Academy required 10 hard questions per lesson eight years ago; now it's four easy ones — it had to drop to get school adoption. IXL can do a SmartScore of 100, which is a real mastery standard, but their site coaches you to stop at 70. Joe was on a call with a YC-funded AI tutor whose founder proudly showed that teachers can adjust the mastery threshold; Joe told him the product won't work. **The one third-party app that holds a 90% gate is eGUMPP** — which Joe credits for Alpha kids' grammar and language, and which kids consider the worst app they have. That tension is the point.

Math Academy does not enforce mastery — two reps can carry you through a lesson. It works for 8th–12th because worked examples dominate once you hit algebra (this is why Alpha's SAT math scores are high). It fails 4th–7th, where reps matter more. IXL at SmartScore 100 would beat Math Academy for 4th–7th; Math Academy beats IXL at algebra and up because IXL lacks worked examples. Hence this year's supplements and an Alpha-built 3rd/4th grade math app with **initial mastery** included.

Failure points, level by level

MAP is not a mastery test. It's relative, it stops asking once you're missing half, and Alpha places students at MAP minus two grade levels because that's roughly where 90% mastery actually sits. Its value is entirely for parents: "twice as much" is comprehensible in a way that "70th to 82nd percentile" isn't, and 2X is measured against kids like yours.

Kids game the baseline. Some tank the fall MAP to set an easy floor, then hit 2X on paper while their year-over-year growth is under 1X. A fix is coming and is deliberately not being announced, because students will find it. One option under consideration: base 2X on the spring test instead.

Motivation on test day is worth a full grade level. Offering Chick-fil-A for matching or beating a prior score moves Alpha students by roughly one grade level. Joe cited the study where paying students *after* the test raises US scores dramatically and Asian scores not at all.

State tests vary wildly. Alaska correlates ~99.8% with MAP; Texas is the worst at ~75% and roughly 19th best as a test overall (MCAS is the best). And no state enforces mastery — Texas treats 50–70% as knowing the material. Alpha's gate is **90%**; the academic team would prefer 95% but that breaks motivation.

Question banks get burned. Kids learn to fail three tests in a row to see the whole bank, then pass. AI now generates effectively infinite question banks, which solves it.

Easy questions are a failure point. Math Academy's SAT supplement capped Alpha students at 650. Alpha sent the data, Math Academy pulled the product for a year, made the questions harder, and kids now reach 790.

New this year: unit quizzes. A grade level is too big a chunk between mastery gates, so groups of lessons will now have AI-generated unit quizzes — the AP model pushed down into the rest of the curriculum.

The metric: first-time right

Borrowed from manufacturing — this is **rework**. If a kid can't clear a mastery gate on the first attempt, that's a failure of the lesson, not the student. Three years ago the doom loop was tolerated because there was no better fix; now the fix goes into the lesson.

Doom loop diagnosis, in order: (1) Is the student motivated? If not, nothing else matters — they aren't engaging with the app at all. (2) Are they missing a prereq? (3) Does the lesson suck?

Alpha Read changes for this year, both classed as Alpha's fault, not the student's: 5% of questions were unfair (two equally defensible answers — the top parent complaint) and have been rewritten; 20% of articles had above-grade-level vocabulary and were rewritten down. The fix is fair questions, **not a lower standard.** Note the leverage: a 5% bad-question rate matters far more than 5% because a single miss inside a string drops you below threshold and kills motivation.

XP is both the motivation loop and the mastery loop. That's the permanent tension. When kids complain, Joe's read is that it means one of three things is broken on Alpha's side — bad lesson, unfair questions, or wrong placement — and all three are fixable.

2X gets *easier* as kids get older

Counterintuitive but load-bearing. First grade is where American students learn the most, so 2X in K–1 means genuinely covering two grade levels and is the hardest case. By 8th grade the median student isn't even at grade level, so 2X requires well under a full grade level — which is why high school runs ~6X and Santa Barbara middle school hit 5X. Even the top 1% aren't learning a full grade level by middle school, because they still have holes.

This is also why some students hole-fill without completing their grade level and still clear 2X.

The advanced-student problem, explained. K–8 runs Common Core and state tests. High school switches entirely to SAT and AP, because that's what colleges care about. **AP Calc BC is a 75-hour course; 7th grade math is 22 hours.** A middle schooler on a 25-minute math block cannot get through AP curriculum — high school runs 3 hours for exactly this reason. The fix in progress: build **non-AP high school grade-level curriculum** for math and science, at less than half the hours (high school bio vs. AP bio), so accelerated middle schoolers have somewhere to go. Reading and language already work this way — 4th and 5th graders place out of 10th–12th grade material because it's grade-level, not AP.

If you have a top-1% middle schooler who can't hit 2X, flag them for an individualized plan. It may mean 45 minutes of math instead of 25.

AlphaTest goes national

Rather than leaning on Texas STAAR, tests are now generated from the **union of every state's standards**, taking the hardest version of each and dropping transient knowledge (who your governor is). Massachusetts' Boston Tea Party, Florida's Trail of Tears, Texas' Alamo — students get all of it. "The alpha test will become the hardest test you can take in America." Context: most states take NAEP and water it down, pushing a 4th grade NAEP standard to 5th or 6th so their pass rates look better.

Andy Montgomery's additions: 3rd–8th math STAAR tests are being made more rigorous, because kids were passing them without the corresponding MAP gain. AP prerequisite tests are being added for Physics and Chemistry, with Biology in progress. The AP Lang problem is different — it's not missing prereqs, it's that AP Lang requires analysis vocabulary that isn't taught anywhere in 8th–10th grade, so direct instruction on those word groups is being built.

The middle school bottleneck: aggregation of holes

Chemistry is hard because chemistry is algebra with word problems — if you've mastered algebra, the incremental chemistry knowledge is small. Algebra is predicted by fast fraction manipulation. Miss 17% of fractions, then 8th grade algebra, then chemistry, and it compounds. Swiss cheese, or Jenga: how many pieces can you pull before it collapses, and where on the curve does it fall over?

"If you wanted to fix education in America, the single best thing you could do is hold kids back who can't read in first grade."

The public-school version of this problem: districts are legally required to deliver grade-level content to a 7th grader who can't read. Alpha's charter/public work (Paige's HISD system) has to show the student attempted 7th grade material and failed twice before remediation is legally permitted — then the system routes them to the 3rd grade lessons they actually need.

The life skills parallel: the same structure, but harder, because there's no app that drops a student four grade levels. Alpha High's boot camp is remediating "receive feedback without crying" — a *kindergarten* alpha test — for 9th grade transfers. Every level has a version of this transfer problem.

Admissions and parent alignment

The room's diagnosis: parents don't always disclose IEPs, 504s, or behavior histories, and crises surface after enrollment. Suggestion made and endorsed — as a private school Alpha can ask for anything, so **request records directly from the previous school with parent consent**, which is routine practice between schools and routes around both concealment and parents who genuinely don't know. Also proposed: level-by-level rubrics defining "not-yet" behaviors, so the bar is explicit for guides and parents.

Shadow days: one day is the honeymoon. Proposal to extend to two days or a week. Joe's answer: *"If the only way we can deliver three commitments is to have everybody do a week, we do a week."*

The campus-shopping problem. Some campuses say no 50% of the time; others almost never. That's not a campus standard problem, it's a missing **Alpha standard**. As soon as a weakness exists, parents find it and route around it — "that's the easy campus, shadow there." Same as kids finding the vocab app that gives cheap XP. **"If we have a really easy campus, everybody's going to do it. That's our fault, not theirs."** A "no" should also hold across campuses and across time.

Why parent alignment is a delivery issue, not a sales issue. Alpha is a package — you can't take the workshops and reject the high standards, because the standards are what make the rest work. If a family is misaligned, the three commitments cannot be delivered to that kid, and the failure traces back to admissions. That's the origin of the parent tenets and the reason the Dean of Parents role exists. Joe's example: a New York group of preschool/kindergarten families whose position amounts to *the building is the most important part of Alpha* — a real preference, just not this school's. The Sports Academy Online version: parents who want the sports and don't want the 2X. They can't come.

Assorted answers

- **Show your work** is the biggest failure mode for GT kids — they do it in their heads until working memory runs out, then fall off. Automating enforcement isn't solved yet; Scribble is the model, and the **Focus Pod** hardware has two non-negotiables: a camera by virtue of being an enclosed pod, and a digital notepad that compiles student notes.
- **StudyFilm is mandatory for online** — "there is no online without StudyFilm this school year." At physical campuses it'll be optional, likely enabled for students below 2X.
- **Alpha Anywhere** is struggling to deliver 2X and is being reworked. Alpha will not sell 2-hour learning standalone if it doesn't deliver 2X. Sports Academy Online works because the sports motivation carries the academics — students scan a QR code proving their two hours before the gym lets them in.
- **Last year's aggregate was 2.6X.** Under it: 9% of kids spent more than two hours and learned less than 2X (the failures worth mining), and ~18% spent more than two hours and got about 3.3X (the system working as intended for kids catching up).

Closing

Alpha starts the year with roughly **2,000 kids in physical schools** and capacity building toward 7,000–10,000. Coming: the Fortnite-based learning game later this year, and a **$200B voucher program next year** that makes online scale financially possible if StudyFilm solves cheating.

Joe's stated priority list, with everything else a distant second: **deliver the three commitments to every kid at the Alpha physical schools.** That's what the world is watching, and delivering it buys time for every other model to work. The failure case he named: "one year from now, we broke at 50."

He closed by saying that if parents could just sit in the building for these two weeks and watch, they'd all want to send their kids here — and that the best birthday present was getting to spend it on this.